

$$\int_a^b f(x) dg(x)$$

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

Chapter 6: The Riemann–Stieltjes Integral

Section: Definition and Existence of the Integral

Based on Walter Rudin's *Principles of Mathematical Analysis*

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Outline

$$\int_a^b f(x) dg(x)$$

Motivation

The Riemann Integral

The Riemann–Stieltjes Integral

Refinements

The Cauchy Criterion for Integrability

Which Functions Are Integrable?

Summary



$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

$$\int_a^b f(x) dg(x)$$

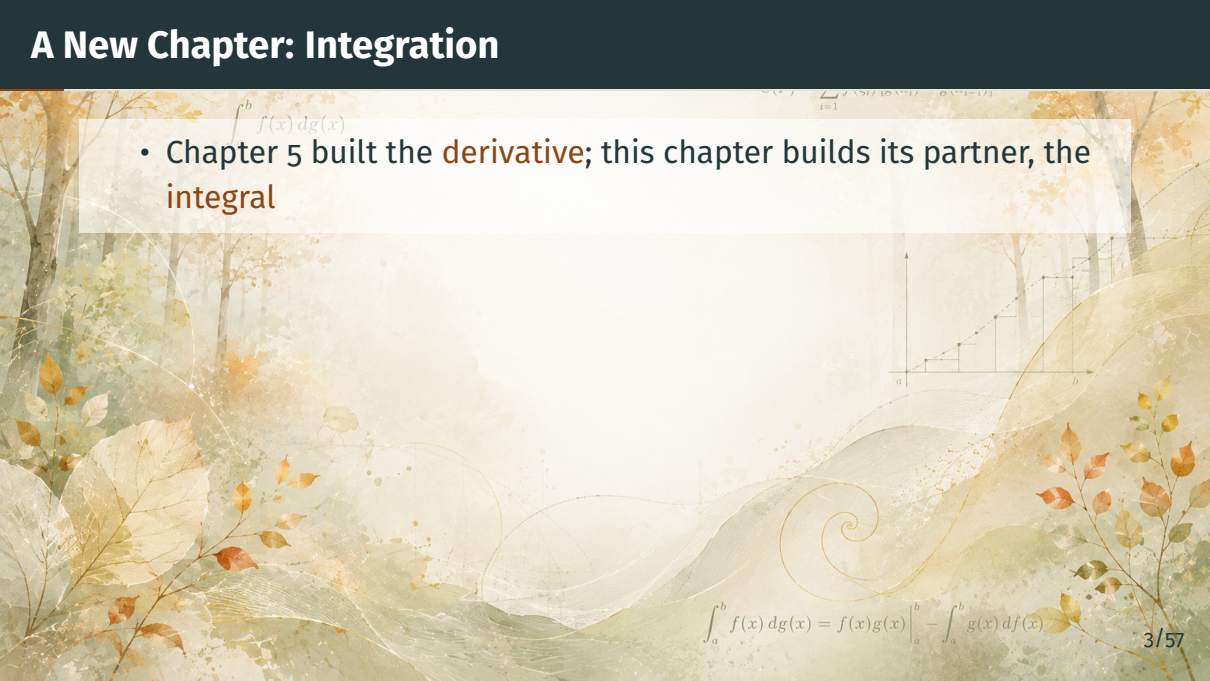
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Motivation



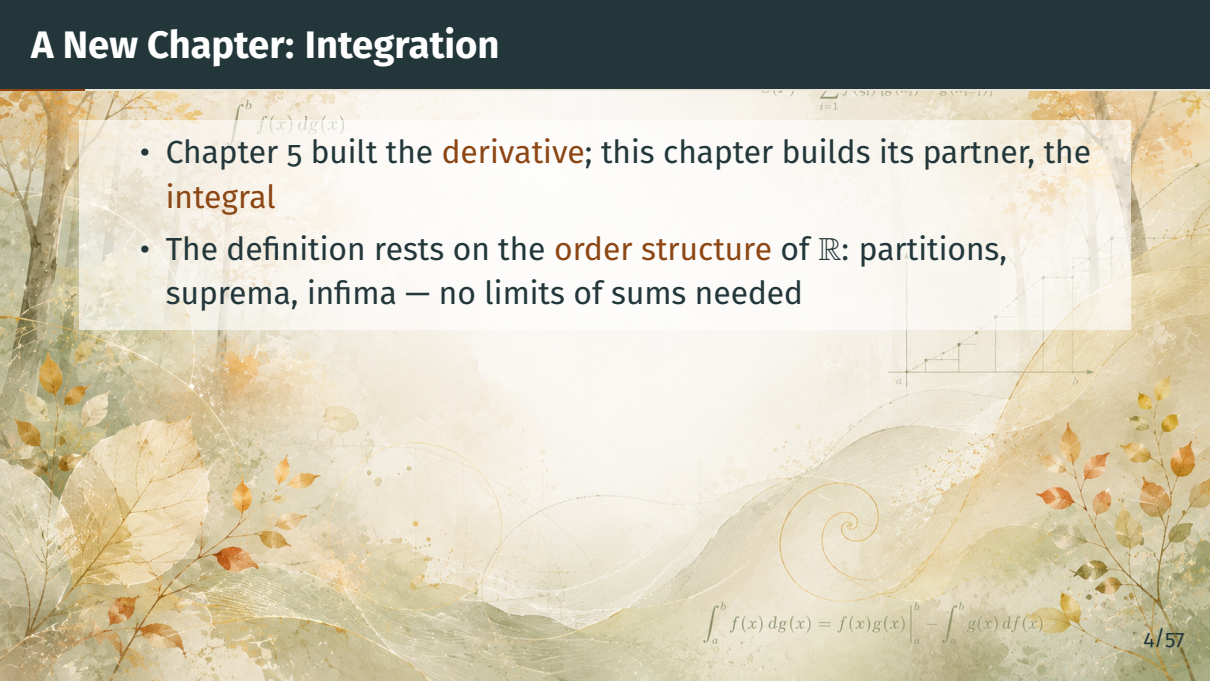
$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

A New Chapter: Integration

- 
- The background features a soft, painterly illustration of a forest scene with trees and autumn leaves in shades of yellow, orange, and brown. Overlaid on this is a semi-transparent white rectangular box containing text. To the right of the text box, a small graph is visible, showing a curve with a step function approximation (Riemann sum) and a smooth curve passing through the steps. The graph has axes and labels 'a' and 'b' on the x-axis.
- Chapter 5 built the **derivative**; this chapter builds its partner, the **integral**

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

A New Chapter: Integration

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- Chapter 5 built the **derivative**; this chapter builds its partner, the **integral**
 - The definition rests on the **order structure** of $\mathbb{R}$: partitions, suprema, infima — no limits of sums needed

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- We generalize immediately: integrate f against an increasing **weight** α

$$\int_a^b f \, dx \quad \rightsquigarrow \quad \int_a^b f \, d\alpha \quad (\text{Riemann-Stieltjes})$$

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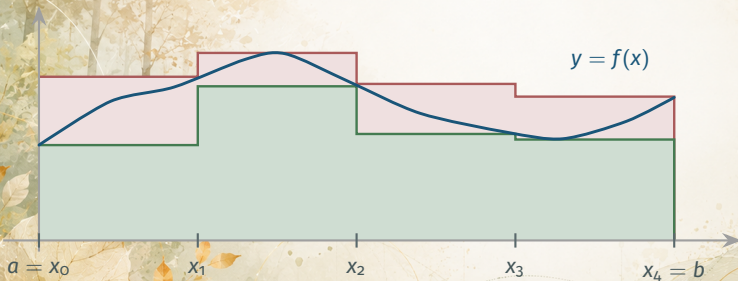
$$\int_a^b f \, dx \quad \rightsquigarrow \quad \int_a^b f \, d\alpha \quad (\text{Riemann-Stieltjes})$$

- Central question: **which functions are integrable?**

$$\int_a^b f(x) \, dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) \, df(x)$$

The Picture: Trapping the Area

$$\int_a^b f(x) dg(x)$$



- lower sum $L(P, f)$
- upper sum $U(P, f)$
- red band = the gap $U - L$

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

$$\int_a^b f(x) dg(x)$$

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

The Riemann Integral



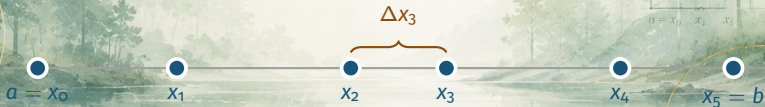
Definition 6.1: Partitions

Definition 6.1 (partition)

A **partition** P of $[a, b]$ is a finite set of points $x_0, x_1, \dots, x_n$ with

$$a = x_0 \leq x_1 \leq \dots \leq x_{n-1} \leq x_n = b.$$

We write $\Delta x_i = x_i - x_{i-1}$ ($i = 1, \dots, n$).



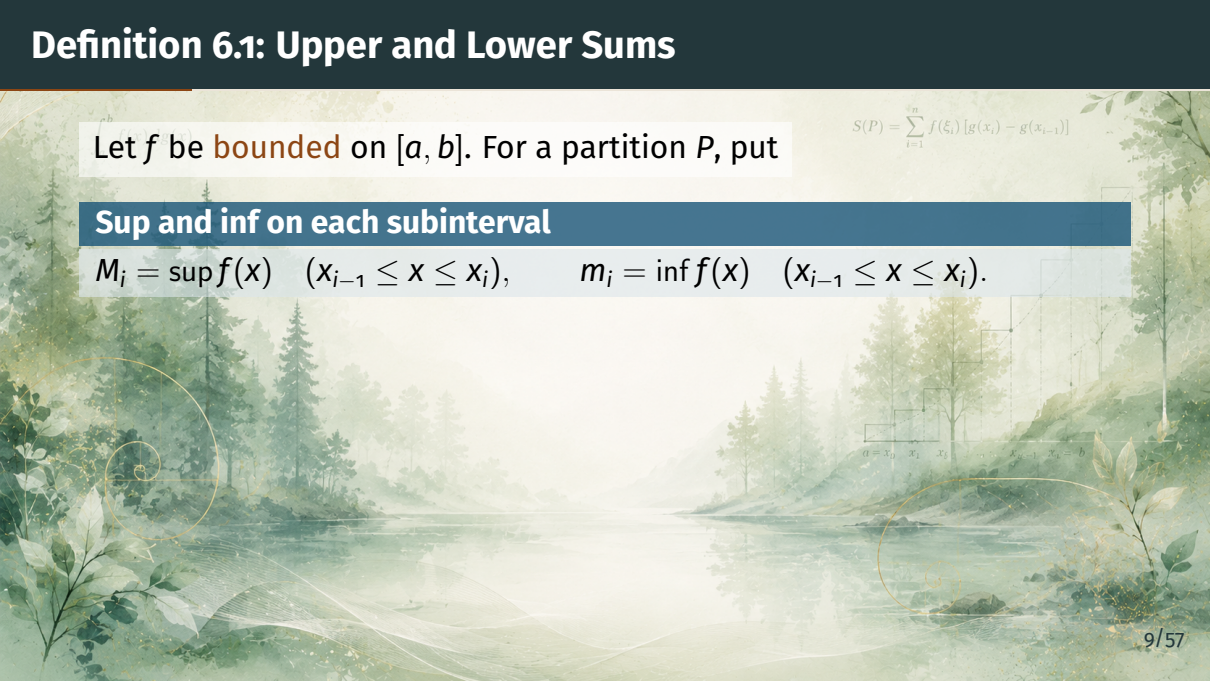
Definition 6.1: Upper and Lower Sums

Let f be **bounded** on $[a, b]$. For a partition P , put

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

Sup and inf on each subinterval

$$M_i = \sup f(x) \quad (x_{i-1} \leq x \leq x_i), \quad m_i = \inf f(x) \quad (x_{i-1} \leq x \leq x_i).$$


$$a = x_0, \quad x_1, \quad x_2, \quad \dots, \quad x_{n-1}, \quad x_n = b$$

Definition 6.1: Upper and Lower Sums

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Upper and lower sums

$$U(P, f) = \sum_{i=1}^n M_i \Delta x_i, \quad L(P, f) = \sum_{i=1}^n m_i \Delta x_i.$$

Always $L(P, f) \leq U(P, f)$, since $m_i \leq M_i$.

Definition 6.1: Upper and Lower Integrals

Upper and lower Riemann integrals

$$\overline{\int_a^b} f dx = \inf_P U(P, f), \quad (1) \quad \underline{\int_a^b} f dx = \sup_P L(P, f), \quad (2)$$

the inf and sup taken over **all** partitions P of $[a, b]$.

Definition 6.1: Upper and Lower Integrals

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the inf and sup taken over **all** partitions P of $[a, b]$.

The Riemann integral

If (1) = (2), we say f is **Riemann-integrable**, write $f \in \mathcal{R}$, and denote the common value by

$$\int_a^b f \, dx, \quad (3) \qquad \text{or} \qquad \int_a^b f(x) \, dx. \quad (4)$$

Upper and Lower Integrals Always Exist

Since f is bounded, there are numbers m, M with $m \leq f(x) \leq M$ on $[a, b]$. Hence for **every** P :

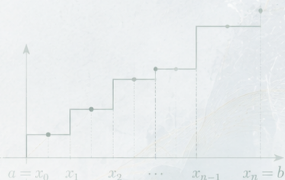
$$m(b-a) \leq L(P, f) \leq U(P, f) \leq M(b-a).$$

Consequence

The numbers $L(P, f), U(P, f)$ form a **bounded set**: the upper and lower integrals are **defined for every bounded f** . Their *equality* — integrability — is the delicate question.

$$\int_a^b f(x) dg(x)$$

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$



The Riemann–Stieltjes Integral

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

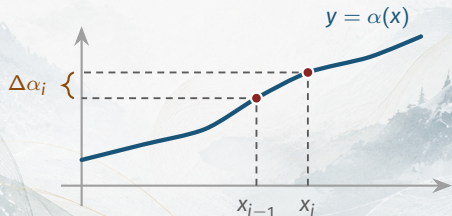
Definition 6.2: An Increasing Weight α

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

Definition 6.2 (setup)

Let α be **monotonically increasing** on $[a, b]$. Since $\alpha(a), \alpha(b)$ are finite, α is bounded. For each partition P write

$$\Delta\alpha_i = \alpha(x_i) - \alpha(x_{i-1}) \geq 0.$$



$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Definition 6.2: Weighted Sums and Integrals

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

Weighted upper and lower sums

For any real f bounded on $[a, b]$ (with M_i, m_i as in Def. 6.1):

$$U(P, f, \alpha) = \sum_{i=1}^n M_i \Delta \alpha_i, \quad L(P, f, \alpha) = \sum_{i=1}^n m_i \Delta \alpha_i,$$

$$\overline{\int_a^b} f d\alpha = \inf_P U(P, f, \alpha), \quad (5)$$

$$\underline{\int_a^b} f d\alpha = \sup_P L(P, f, \alpha). \quad (6)$$

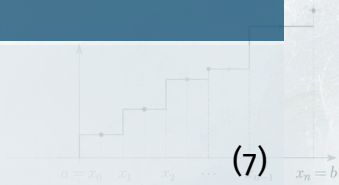
$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

The Riemann–Stieltjes Integral

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

The integral of f with respect to α

If (5) = (6), we denote the common value by

$$\int_a^b f d\alpha \quad (7)$$


or

$$\int_a^b f(x) d\alpha(x). \quad (8)$$

This is the **Riemann–Stieltjes integral** of f with respect to α over $[a, b]$; we say f is integrable with respect to α and write $f \in \mathcal{R}(\alpha)$.

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Two Remarks on the Definition

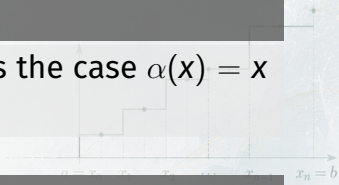
$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

The Riemann integral is a special case

Taking $\alpha(x) = x$ gives $\Delta\alpha_i = \Delta x_i$: Riemann integral is the case $\alpha(x) = x$ of the Riemann–Stieltjes integral.

α can be wild

In general, α **need not even be continuous** — step functions are allowed (and will be important!).



$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

The Variable of Integration Is a Dummy

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

Notation

We prefer (7) to (8): the letter x in (8) **adds nothing**. For instance, (8) is the same as

$$\int_a^b f(y) d\alpha(y).$$

The integral depends on f , α , a , b — *not* on the variable of integration. It is analogous to the index of summation:

$$\sum_{i=1}^n c_i \quad \text{and} \quad \sum_{k=1}^n c_k \quad \text{both mean} \quad c_1 + c_2 + \cdots + c_n.$$

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Standing Assumptions

$$S(P) = \sum_{i=1} f(\xi_i) [g(x_i) - g(x_{i-1})]$$

For the rest of this chapter (unless said otherwise)

- f is **real and bounded** on $[a, b]$
- α is **monotonically increasing** on $[a, b]$
- When no confusion can arise, we write $\int$ for $\int_a^b$



Goal: decide **when** $f \in \mathcal{R}(\alpha)$.

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

$$\int_a^b f(x) dg(x)$$

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

Refinements



$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Definition 6.3: Refinements

Definition 6.3

P^* is a **refinement** of P if $P^* \supset P$ (every point of P is a point of P^*). Given P_1, P_2 , their **common refinement** is $P^* = P_1 \cup P_2$.



$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Theorem 6.4: Refining Improves Both Sums

Theorem 6.4

If P^* is a refinement of P , then

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

$$L(P, f, \alpha) \leq L(P^*, f, \alpha) \quad (9)$$

and

$$U(P^*, f, \alpha) \leq U(P, f, \alpha). \quad (10)$$

Key idea: adding points can only **tighten** the trap: lower sums go up, upper sums come down.

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Proof of Theorem 6.4 — Step 1: One Extra Point

Step 1: Setup

Suppose first P^* has **just one more point** x^* than P , with $x_{i-1} < x^* < x_i$.

Put $S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$

$$w_1 = \inf f(x) \quad (x_{i-1} \leq x \leq x^*), \quad w_2 = \inf f(x) \quad (x^* \leq x \leq x_i).$$

Then $w_1 \geq m_i$ and $w_2 \geq m_i$, where $m_i = \inf f(x)$ on $[x_{i-1}, x_i]$.

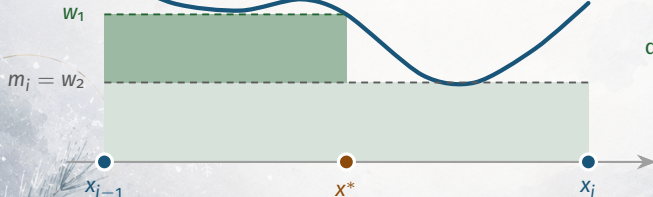
Why: an inf over a **smaller** set can only be **larger**.

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

The Picture: Why Lower Sums Rise

$$\int_a^b f(x) dg(x)$$

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$



darker block = gain in L^b

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Proof of Theorem 6.4 — Step 2: Compute the Gain

Step 2: The gain is nonnegative

$$\begin{aligned} S(P) &= \sum_{i=1}^n [L(P^*, f, \alpha) - L(P, f, \alpha)] = w_1[\alpha(x^*) - \alpha(x_{i-1})] + w_2[\alpha(x_i) - \alpha(x^*)] \\ &\quad - m_i[\alpha(x_i) - \alpha(x_{i-1})] \\ &= (w_1 - m_i)[\alpha(x^*) - \alpha(x_{i-1})] \\ &\quad + (w_2 - m_i)[\alpha(x_i) - \alpha(x^*)] \geq 0. \end{aligned}$$

Each factor is ≥ 0 : $w_1, w_2 \geq m_i$ and α is **increasing**.

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Proof of Theorem 6.4 — Step 3: Iterate

Step 3: Finish

If P^* contains k points more than P , repeat the argument k times: this proves (9). The proof of (10) is **analogous** (with sups in place of infs). $\square$

Technique: reduce to the **one-point** case, then iterate.

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Theorem 6.5: Lower Never Exceeds Upper

Theorem 6.5

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

$$\underline{\int_a^b f d\alpha} \leq \overline{\int_a^b f d\alpha}.$$



Not obvious! (2) and (5) involve **different** partitions: we must compare $L(P_1, f, \alpha)$ with $U(P_2, f, \alpha)$ for **arbitrary** P_1, P_2 .

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Proof of Theorem 6.5 — The Chain

Proof

Let P^* be the common refinement of P_1 and P_2 . By Theorem 6.4,

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

$$L(P_1, f, \alpha) \leq L(P^*, f, \alpha) \leq U(P^*, f, \alpha) \leq U(P_2, f, \alpha).$$

Hence

$$L(P_1, f, \alpha) \leq U(P_2, f, \alpha). \quad (11)$$

Fix P_2 , take sup over all P_1 :

$$\underline{\int} f d\alpha \leq U(P_2, f, \alpha). \quad (12)$$

Take inf over all P_2 in (12): $\underline{\int} f d\alpha \leq \overline{\int} f d\alpha.$

□

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

$$\int_a^b f(x) dg(x)$$

The Cauchy Criterion for Integrability



$$\sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

Theorem 6.6: The Integrability Criterion

Theorem 6.6

$f \in \mathcal{R}(\alpha)$ on $[a, b]$ **if and only if** for every $\varepsilon > 0$ there exists a partition P such that

$$U(P, f, \alpha) - L(P, f, \alpha) < \varepsilon. \quad (13)$$



Key idea: no need to compute either integral — just make the **red band** thin with *one* good partition.

$$\sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

Proof of Theorem 6.6 — “If”: The Squeeze

(13) for every $\varepsilon \implies f \in \mathcal{R}(\alpha)$

For every P :

$$L(P, f, \alpha) \leq \int f d\alpha \leq \overline{\int} f d\alpha \leq U(P, f, \alpha).$$

Thus (13) implies

$$0 \leq \overline{\int} f d\alpha - \int f d\alpha < \varepsilon.$$

True for every $\varepsilon > 0 \implies$ the two integrals are **equal**: $f \in \mathcal{R}(\alpha)$.

$$\int_a^b f(x) dg(x)$$



$$\sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

Proof of Theorem 6.6 — “Only If”: Step 1

Step 1: Two good partitions

Suppose $f \in \mathcal{R}(\alpha)$, let $\varepsilon > 0$. Since $\int f d\alpha$ is both $\sup_P L$ and $\inf_P U$, there exist partitions P_1, P_2 with

$$U(P_2, f, \alpha) - \int f d\alpha < \frac{\varepsilon}{2}, \quad (14)$$

$$\int f d\alpha - L(P_1, f, \alpha) < \frac{\varepsilon}{2}. \quad (15)$$

Problem: two different partitions. **Fix:** their common refinement.

Proof of Theorem 6.6 — “Only If”: Step 2

Step 2: Merge and chain

Let P be the common refinement of P_1 and P_2 . Theorem 6.4 with (14) and (15) gives

$$U(P, f, \alpha) \leq U(P_2, f, \alpha) < \int f d\alpha + \frac{\varepsilon}{2} < L(P_1, f, \alpha) + \varepsilon \leq L(P, f, \alpha) + \varepsilon,$$

so (13) holds for this P . □

$$\sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

Theorem 6.7: Working with the Criterion

Theorem 6.7(a)

If (13) holds for some P and some ε , then (13) holds (with the same ε) for **every refinement** of P .

$$\int_a^b f(x) dg(x)$$



$$\sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

Theorem 6.7: Working with the Criterion

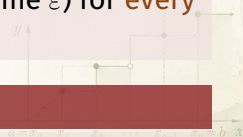
Theorem 6.7(a)

If (13) holds for some P and some ε , then (13) holds (with the same ε) for **every refinement** of P .

Theorem 6.7(b)

If (13) holds for $P = \{x_0, \dots, x_n\}$ and s_i, t_i are **arbitrary points** in $[x_{i-1}, x_i]$, then $\sum_{i=1}^n |f(s_i) - f(t_i)| \Delta\alpha_i < \varepsilon$.

$$\int_a^b f(x) dg(x)$$



$$\sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

Theorem 6.7: Working with the Criterion

Theorem 6.7(a)

If (13) holds for some P and some ε , then (13) holds (with the same ε) for **every refinement** of P .

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If (13) holds for $P = \{x_0, \dots, x_n\}$ and s_i, t_i are **arbitrary points** in $[x_{i-1}, x_i]$, then $\sum_{i=1}^n |f(s_i) - f(t_i)| \Delta\alpha_i < \varepsilon$.

Theorem 6.7(c)

If $f \in \mathcal{R}(\alpha)$ and the hypotheses of (b) hold, then $|\sum_{i=1}^n f(t_i) \Delta\alpha_i - \int_a^b f d\alpha| < \varepsilon$.

Proof of Theorem 6.7 — Parts (a) and (b)

(a)

Refine P : by Theorem 6.4, U does not increase and L does not decrease, so $U - L$ does not increase. $\Rightarrow$ (13) persists.

(b)

Both $f(s_i)$ and $f(t_i)$ lie in $[m_i, M_i]$, so $|f(s_i) - f(t_i)| \leq M_i - m_i$. Hence

$$\sum_{i=1}^n |f(s_i) - f(t_i)| \Delta\alpha_i \leq U(P, f, \alpha) - L(P, f, \alpha) < \varepsilon.$$

Proof of Theorem 6.7 — Part (c)

(c)

The obvious inequalities

$$L(P, f, \alpha) \leq \sum f(t_i) \Delta \alpha_i \leq U(P, f, \alpha)$$

and

$$L(P, f, \alpha) \leq \int f d\alpha \leq U(P, f, \alpha)$$

prove (c): both numbers lie in an interval of length $< \varepsilon$. □

Interpretation: Riemann–Stieltjes **sums** converge to the **integral** as partitions get good.

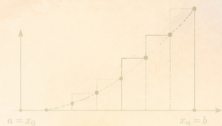
$$\int_a^b f(x) dg(x)$$



$$\sum_{i=1}^n f(t_i) [g(x_i) - g(x_{i-1})]$$

$$\int_a^b f(x) dg(x)$$

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$



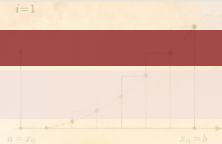
Which Functions Are Integrable?

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Theorem 6.8: Continuous Functions

Theorem 6.8

If f is **continuous** on $[a, b]$, then $f \in \mathcal{R}(\alpha)$ on $[a, b]$.



Strategy:

- continuity on a compact interval $\Rightarrow$ **uniform** continuity
- uniform continuity $\Rightarrow$ small oscillation $M_i - m_i$ on **every** fine subinterval
- then $U - L$ telescopes to something small

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Proof of Theorem 6.8 — Step 1: Choose η, δ

$i=1$

Step 1: Calibrate

Let $\varepsilon > 0$. Choose $\eta > 0$ so that

$$[\alpha(b) - \alpha(a)] \eta < \varepsilon.$$

f is **uniformly continuous** (Theorem 4.19): there is $\delta > 0$ with

$$|f(x) - f(t)| < \eta \tag{16}$$

whenever $x, t \in [a, b]$, $|x - t| < \delta$.

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Proof of Theorem 6.8 — Step 2: Telescope

$i=1$



Step 2: A fine partition works

Take any P with $\Delta x_i < \delta$ for all i . Then (16) gives

$$M_i - m_i \leq \eta \quad (i = 1, \dots, n), \quad (17)$$

and therefore

$$\begin{aligned} U(P, f, \alpha) - L(P, f, \alpha) &= \sum_{i=1}^n (M_i - m_i) \Delta \alpha_i \leq \eta \sum_{i=1}^n \Delta \alpha_i \\ &= \eta [\alpha(b) - \alpha(a)] < \varepsilon. \end{aligned}$$

By Theorem 6.6, $f \in \mathcal{R}(\alpha)$.



$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Theorem 6.9: Monotonic Functions

Theorem 6.9

If f is **monotonic** on $[a, b]$ and α is **continuous** on $[a, b]$, then $f \in \mathcal{R}(\alpha)$.
(As always, α is monotonic too.)

Note the trade: f may *jump*, but now α must be continuous.

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Proof of Theorem 6.9 — Step 1: Slice α Evenly

$i=1$

Step 1: Equal α -increments

Let $\varepsilon > 0$, n a positive integer. Choose P with

$$\Delta\alpha_i = \frac{\alpha(b) - \alpha(a)}{n} \quad (i = 1, \dots, n).$$

Possible because α is **continuous** (Intermediate Value Theorem 4.23).



$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Proof of Theorem 6.9 — Step 2: Telescope in f

$i=1$

Step 2: Monotonicity locates sup and inf

Say f is increasing (the other case is analogous). Then

$$M_i = f(x_i), \quad m_i = f(x_{i-1}),$$

so

$$\begin{aligned} U(P, f, \alpha) - L(P, f, \alpha) &= \frac{\alpha(b) - \alpha(a)}{n} \sum_{i=1}^n [f(x_i) - f(x_{i-1})] \\ &= \frac{\alpha(b) - \alpha(a)}{n} \cdot [f(b) - f(a)] < \varepsilon \end{aligned}$$

for n large. By Theorem 6.6, $f \in \mathcal{R}(\alpha)$.



$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Theorem 6.10: Finitely Many Discontinuities

$\epsilon=1$

Theorem 6.10

Suppose f is bounded on $[a, b]$ with only **finitely many** points of discontinuity, and α is **continuous at every point where f is discontinuous**. Then $f \in \mathcal{R}(\alpha)$.

Strategy: **quarantine** the bad points in tiny intervals where α barely grows; on the rest, f is uniformly continuous.

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Proof of Theorem 6.10 — Step 1: Quarantine

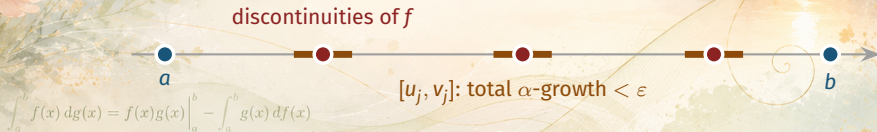
$\epsilon = 1$

Step 1: Cover the bad set

Let $\epsilon > 0$, $M = \sup |f(x)|$, and let E be the (finite) set of discontinuities. Since α is continuous at each point of E , cover E by **finitely many disjoint** intervals $[u_j, v_j] \subset [a, b]$ with

$$\sum_j [\alpha(v_j) - \alpha(u_j)] < \epsilon,$$

placed so every point of $E \cap (a, b)$ is **interior** to some $[u_j, v_j]$.



Proof of Theorem 6.10 — Step 2: The Good Set Is Compact

$\epsilon = 1$

Step 2: Uniform continuity off the bad intervals

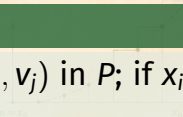
Remove the open segments (u_j, v_j) from $[a, b]$: the remaining set K is **compact**, and f is continuous on K , hence **uniformly continuous**: there is $\delta > 0$ with

$$|f(s) - f(t)| < \epsilon \quad \text{if } s, t \in K, \quad |s - t| < \delta.$$

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Proof of Theorem 6.10 — Step 3: Build P and Estimate

$i=1$



Step 3: The partition

Form P : every $u_j \in P$, every $v_j \in P$, **no point** of any (u_j, v_j) in P ; if x_{i-1} is not a u_j , then $\Delta x_i < \delta$. Then

$$M_i - m_i \leq 2M \text{ always,} \quad M_i - m_i \leq \varepsilon \text{ unless } x_{i-1} \text{ is some } u_j.$$

As in Theorem 6.8:

$$U(P, f, \alpha) - L(P, f, \alpha) \leq [\alpha(b) - \alpha(a)]\varepsilon + 2M\varepsilon.$$

ε arbitrary $\Rightarrow f \in \mathcal{R}(\alpha)$ by Theorem 6.6.



$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

A Warning: Shared Jumps Destroy Integrability

$t=1$

Note (after Theorem 6.10)

If f and α have a **common** point of discontinuity, then f need **not** be in $\mathcal{R}(\alpha)$. (Exercise 3 shows this.)

Moral: the hypotheses of 6.8–6.10 are a balancing act between f and α :
at **each point**, at **most one of them** may jump.

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Theorem 6.11: Composition with Continuous ϕ

Theorem 6.11

Suppose $f \in \mathcal{R}(\alpha)$ on $[a, b]$, $m \leq f \leq M$, ϕ is **continuous** on $[m, M]$, and $h(x) = \phi(f(x))$ on $[a, b]$. Then $h \in \mathcal{R}(\alpha)$ on $[a, b]$.

Examples to come: $\phi(t) = t^2$ and $\phi(t) = |t| \Rightarrow f^2, |f| \in \mathcal{R}(\alpha)$ (used to prove $fg \in \mathcal{R}(\alpha)$ in the next section).

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Proof of Theorem 6.11 — Strategy

Goal: make $U(P, h, \alpha) - L(P, h, \alpha)$ small, knowing that this works for f .

Approach:

1. ϕ is **uniformly continuous**: small input gaps $\Rightarrow$ small output gaps
2. Take P with $U(P, f, \alpha) - L(P, f, \alpha) < \delta^2$ (note: δ **squared**!)
3. Split indices: class A (small oscillation of f), class B (large oscillation)
4. On A: ϕ keeps oscillations small. On B: oscillation $\leq 2K$, but B carries **little α -weight**

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Proof of Theorem 6.11 — Step 1: Calibrate and Partition

$t=1$

Step 1: Choices

Choose $\varepsilon > 0$. Uniform continuity of ϕ on $[m, M]$ gives $\delta > 0$ with $\delta < \varepsilon$ and

$$|\phi(s) - \phi(t)| < \varepsilon \quad \text{if } |s - t| \leq \delta, \quad s, t \in [m, M].$$

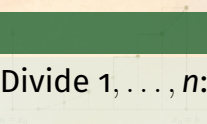
Since $f \in \mathcal{R}(\alpha)$, there is a partition P with

$$U(P, f, \alpha) - L(P, f, \alpha) < \delta^2. \quad (18)$$

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Proof of Theorem 6.11 — Step 2: Two Classes

$i=1$



Step 2: Classes A and B

Let M_i^*, m_i^* be the sup/inf of h (with M_i, m_i those of f). Divide $1, \dots, n$:

$$i \in A \text{ if } M_i - m_i < \delta, \quad i \in B \text{ if } M_i - m_i \geq \delta.$$

- $i \in A$: $M_i^* - m_i^* \leq \varepsilon$ (choice of δ)
- $i \in B$: $M_i^* - m_i^* \leq 2K$, where $K = \sup_{[m, M]} |\phi(t)|$

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Proof of Theorem 6.11 — Step 3: Class B Has Small Weight

$i=1$

Step 3: The δ^2 trick

By (18),

$$\delta \sum_{i \in B} \Delta \alpha_i \leq \sum_{i \in B} (M_i - m_i) \Delta \alpha_i < \delta^2, \quad (19)$$

so that

$$\sum_{i \in B} \Delta \alpha_i < \delta$$

The point of δ^2 : one δ pays for the **height** $2K$, the other for the **width**.

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Proof of Theorem 6.11 — Step 4: Assemble

$i=1$



Step 4: Conclusion

$$\begin{aligned} U(P, h, \alpha) - L(P, h, \alpha) &= \sum_{i \in A} (M_i^* - m_i^*) \Delta \alpha_i + \sum_{i \in B} (M_i^* - m_i^*) \Delta \alpha_i \\ &\leq \varepsilon [\alpha(b) - \alpha(a)] + 2K\delta < \varepsilon [\alpha(b) - \alpha(a) + 2K]. \end{aligned}$$

ε arbitrary $\Rightarrow h \in \mathcal{R}(\alpha)$ by Theorem 6.6. □

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Remark: Just *Which* Functions Are Integrable?

$\epsilon=1$

Remark (after Theorem 6.11)

Theorem 6.11 suggests the question: **just what functions are Riemann-integrable?** The complete answer comes much later, in Theorem 11.33(b) (Lebesgue's criterion).

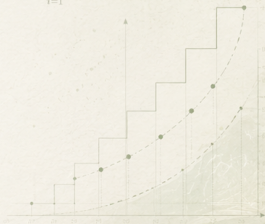
Spoiler: $f \in \mathcal{R} \iff f$ is bounded and continuous **almost everywhere** — but “almost everywhere” needs *measure theory* (Chapter 11).

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

$$\int_a^b f(x) dg(x)$$

Summary

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$



$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Summary

Key takeaways from this section:

- **Two definitions:** upper/lower sums define $\overline{\int} f dx \geq \underline{\int} f dx$; equality = **integrability**. With $\Delta\alpha_i$ for Δx_i : the **Riemann–Stieltjes** integral $\int f d\alpha$

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Summary

Key takeaways from this section:

- **Two definitions:** upper/lower sums define $\overline{\int} f \, dx \geq \underline{\int} f \, dx$; equality = **integrability**. With $\Delta\alpha_i$ for Δx_i : the **Riemann–Stieltjes** integral $\int f \, d\alpha$
- **The machinery:** refinements tighten the trap (6.4), so $\underline{\int} \leq \overline{\int}$ (6.5);
Theorem 6.6: $f \in \mathcal{R}(\alpha) \iff U - L < \varepsilon$ for some P ;
Riemann sums then approximate the integral (6.7)

$$\int_a^b f(x) \, dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) \, df(x)$$

Summary

Key takeaways from this section:

- **Two definitions:** upper/lower sums define $\overline{\int} f \, dx \geq \underline{\int} f \, dx$; equality = **integrability**. With $\Delta\alpha_i$ for Δx_i : the **Riemann–Stieltjes** integral $\int f \, d\alpha$
- **The machinery:** refinements tighten the trap (6.4), so $\underline{\int} \leq \overline{\int}$ (6.5);
Theorem 6.6: $f \in \mathcal{R}(\alpha) \iff U - L < \varepsilon$ for some P ;
Riemann sums then approximate the integral (6.7)
- **Who is integrable:** f continuous (6.8); f monotonic + α continuous (6.9); finitely many discontinuities avoiding those of α (6.10); $\phi \circ f$, ϕ continuous (6.11)

Coming up next: Properties of the Integral — linearity, products, **series as integrals**.

$$\int_a^b f(x) \, dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) \, df(x)$$